

Dr. Sudipta Sarkar

Research Assistant Professor
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Resume website

RESEARCH INTERESTS

My research interests lie in the study of Frames for shift-invariant spaces, situated broadly within the field of Harmonic analysis. Specifically, I focus on the theory of shift-generated frames in the context of locally compact groups, with particular emphasis on nilpotent Lie groups, such as the Heisenberg group. Additionally, I have a keen interest in exploring the Hardy spaces.

JOB EXPERIENCE

- **SRMIST, Chennai** Kattankulathur, India.
Research Assistant Professor December, 2025 - -

EDUCATION AND RESEARCH EXPERIENCE

- **Indian Institute of Technology Guwahati (IIT Guwahati)** Guwahati, India.
Post-Doctoral Fellow December, 2023 - Dec, 2025
- **Stat-Math Unit, Indian Statistical Institute (ISI), Kolkata** Kolkata, India.
Visiting Scientist May, 2023 - November, 2023
- **Indian Institute of Technology Indore (IIT Indore)** Indore, India.
Doctor of Philosophy (Ph.D.) July, 2017 - June, 2023
Thesis Title: Translation Generated Oblique Dual Frames by Actions of Locally Compact Groups
Supervisor: Prof. Niraj Kumar Shukla, Department of Mathematics, IIT Indore
- **Indian Institute of Technology Kanpur (IIT Kanpur)** Kanpur, India.
M.Sc. in Mathematics, CPI 7.5 2014 - 2016
- **Ramakrishna Mission Vidyamandira, Belur Math** Howrah, West Bengal, India.
B.Sc. (Honours) in Mathematics, Marks: 52.50% 2011 - 2014
- **WBCHSE** West Bengal, India.
Higher Secondary in Science, Marks: 80% 2009 - 2011
- **WBBSE** West Bengal, India.
Madhyamik, Marks: 85.25% 2009

PUBLISHED PAPERS

1. **S. Sarkar**, Conditions for frame of translates associated with model sets, *Indian Academy of Sciences. Proc Math Sci*, (2026), accepted.
2. **S. Sarkar, N. K. Shukla**, Shift-Generated dual frames on a class of nilpotent Lie groups, *Rendiconti del Circolo Matematico di Palermo Series 2*, (2026).
3. **S. Sarkar**, Orthogonal Gabor frames for model sets, *Complex Analysis and Operator Theory*, (2025), Vol-19, Paper no. 207.
4. **S. Sarkar and N. K. Shukla**, Characterizations of translation generated dual frames using infimum cosine angle, *Results Math.*, (2025), Paper No. 37.
5. **S. Kalra, S. Sarkar and N. K. Shukla**, An application of the supremum cosine angle between multiplication invariant spaces in $L^2(X; \mathcal{H})$, *Indian Academy of Sciences. Proc Math Sci*, **134**, (2024), Paper no. 15.
6. **S. Sarkar and N. K. Shukla**, Subspace dual and orthogonal frames by action of an abelian group., *J. Pseudo-Differ. Oper. Appl.*, **15**, (2024), Paper no. 32.
7. **S. Sarkar and N. K. Shukla**, Characterizations of extra-invariant spaces under the left translations on a Lie group, *Adv. Oper. Theory*, **8**, (2023), Paper no. 49.
8. **S. Sarkar and N. K. Shukla**, Translation generated oblique dual frames on locally compact groups, *Linear Multilinear Algebra*, **72** (7), (2023), 1188–1219.

SUBMITTED PAPERS/PREPRINTS

1. **S. Sarkar**, Orthogonal frame of translates for model sets, (2025), *submitted to journal*.
2. **S. Sarkar**, Dual frames for subspaces invariant under rigid motions, (2024), *submitted to journal*.
3. **Anjali, A. Chattopadhyay, S. Sarkar, R. K. Srivastava**, Sampling and Interpolation for twisted Fock spaces (2025), *submitted*.

CONFERENCES TALK/ATTENDED

- ICAIPA-2022 (July 18–21, 2022) at IIT Madras, India.
Title of the talk - Multiplication generated duals of a frame.
- 17'th DMHA (January 5– 8, 2022), School of Mathematical Sciences, NISER, India.
Title of the talk - Reproducing formula on Nilpotent Lie group.
- 27'th International Conference of the International Academy of Physical Sciences (CONIAPS XXVII) (October 26th-28th, 2021), Department of Mathematics, Central University of Kerala, India.
Title of the talk - Translation generated duals of a subspace frame.
- 35'th Annual Conference of Ramanujan Mathematical Society (December 28–30, 2020), Department of Mathematics, Central University of Rajasthan.
Title of the talk - MG and TG-duals of a Subspace Frame on Locally Compact Groups.
- DMHA at IIT Guwahati (December 18–21, 2023), IIT Guwahati, India.
- Online International Conference on Computational Harmonic Analysis (September 13–17, 2020).
- DMHA at IISER Bhopal (December 16–19, 2019) at IISER Bhopal, India.
- International Conference of Mathematical Modelling and Scientific Computing (ICMMSC-2018) (July 19–21, 2018) IIT Indore, India.
- Global Initiative of Academic Network (GIAN) course on “Theory and Application of Wavelets and Framelets” (December 28–January 04, 2018) IIT Indore, India.

WORKSHOPS ATTENDED

- **Resource Person** in a Five-Day Online FDP on “Wavelet and its Applications” (April 15–19, 2024), Department of Mathematics, School of Advanced Sciences, Sciences, VIT, Chennai.
- Online short-term course “Sampling theory and its Applications: Signal, Image processing and Data Science” (January 24–29, 2022) , IIT Indore, India.
- Workshop on Geometry of Banach Spaces and its Applications (December 31– January 02, 2022), IIIT Allahabad, India.
- International Workshop on Wavelets and its Applications: Image Processing, Data Science and PDEs (WAIDP-2021) (December 06–10, 2021), Manaba Rachana university, India.
- National Workshop on Topology and Functional Analysis (March 11–15, 2019), LNMIIT Jaipur, Rajasthan, India.
- Inverse problem and Imaging Techniques (February 21–27, 2019), IIT Indore, India.
- Annual Foundation School (AFS-I) (May 7–June 2, 2018), IIT Kanpur, India.

TEACHING EXPERIENCES

- Mathematics-I (MA102) (IIT Guwahati) (Jan–Jul, 2025)
- Mathematics-I (MA101) (IIT Guwahati) (Jul–Dec, 2024)
- Mathematics - III (MA201) (IIT Guwahati) (Jul–Dec, 2024)
- Calculus (MA105) (IIT Indore) (Nov–Feb, 2022)
- Calculus (MA105) (IIT Indore) (Nov–Feb, 2021)
- Complex Analysis and Differential Equations-II (MA203) (IIT Indore) (Jul–Dec, 2020)
- Linear Algebra and Ordinary Differential Equations-I (MA106) (IIT Indore) (Jan–Jul, 2020)
- Calculus (MA105) (IIT Indore) (Jul–Dec, 2019)
- Linear algebra and ODE-I (MA106) (IIT Indore) (Jan–Jul, 2019)
- Complex Analysis and Differential Equations-II (MA203) (IIT Indore) (Jul–Dec, 2018)

SCHOLARSHIP, AWARDS AND FELLOWSHIP

- Qualified **CSIR NET** (Mathematical Sciences) for Junior Research Fellowship in June 2016.

EXTRACURRICULAR ACTIVITIES

- Participation in **Yuva sangam Program**, an initiative by **Prime Minister's Ek Bharat Shreshtha Bharat** programme, with an exposure to north-east state **Manipur** from March 27–April 02, 2023.
- **Student Representative** in Departmental Postgraduate Committee (DPGC) at IIT Indore, India, February 2020–March 2023.
- Member of the Safety Management Committee at IIT Indore, India, July 2019–December 2021.

HOBBIES

- Study of Indian philosophy, in particular non-dualistic Vedanta, reading books, etc.

PERSONAL INFORMATION

- **Date of Birth:** May 26, 1993
- **Nationality:** Indian

REFERENCES

Prof. Niraj Kumar Shukla

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Prof. Biswaranjan Behera

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Prof. Rajesh K. Srivastava

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Prof. Arup Chattopadhyaya

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